Numerical Analysis Statistical Analysis Presentation

By Courtney Sommerville (12693871)

Introduction

- Alcohol is a widely used beverage
- Alcohol is a leading cause for hospital admissions.
- It is estimated in 2017-2018 there were 338 thousand admissions due to alcohol.
- There were also 5,843 deaths due to alcohol.
- This is 6% higher than 2016 and 16% higher than 2007.
- 57% of adults drank up to 14 units of alcohol each week and 24% drank more than this.
- 82% of adults have drank alcohol in the last 12 months
- 49% of adults drink at least once a week.

- The purpose of this assignment is to be able to use R to look at the data given and perform different tests and to determine if variables are statistically different.
- Using the data from health survey for England 2011, I have gathered all the data in the powerpoint I am about to show you.
- I used lots of different analysis during this task, this consisted of statistical difference so I performed a t-test, aswell as finding the values for different questions and also finding the mean, median, mode, and standard deviation and more.
- R is used to perform these tests because its a programming language that allows statistical computing and shows data we are looking for in a large data set.

Total people included in the sample	**10617**
% of people drink alcohol	63.2%
% of women in sample	54.2%
Highest educational level	NVQ4/NVQ5/Degree or equivalent
Divorced	5.59%
Separated	2.1%

So we have a total of 10,617 people in the sample data, 54.2% are female. 63.2% of them drink alcohol. There is also 5.59% divorced and 2.1% separated according to the data. The highest level of education in the sample data is NVQ4/NVQ5/ Degree or equivalent.

Here I calculated the mean, median, mode, minimum and maximum, range and standard deviation of the variables household income, BMI and age at last birthday.
The minimum BMI is 8.34 and the maximum BMI is 65.28. The mean household size is 2.85. The median age at last birthday is 42 and the standard deviation for BMI is 6.138.

	Household size	BMI	Age at last birthday
Mean	2.85	25.92	41.56
Median	3	25.59	42
Mode	2	13.767	42
Minimum	1	8.34	0
Maximum	10	65.28	100
Range	10-1=9	65.28-8.34=56.94	100-0=100
Standard deviation	1.369	6.138	23.832

	Drinking status		Value of the test and p value
Gender	Yes	No	X2 = 114.15 P value = 1.20834e-26
Male	3172	3540	
Female	605	1217	

Null hypothesis : There is no relationship between gender and drinking status
I ran a significance test to find out which gender drinks more alcohol and the P-Value is <0.05 so therefore we can reject the null hypothesis as there is a significant difference between gender and drinking.

- **Null hypothesis : there is no significant difference between drinking and region.**
- **I ran a significance test to find out which region drinks more alcohol and the P-Value is <0.05 so therefore we can reject the null hypothesis as there is a significant difference between drinking and the region they are in.**

	Drinking status		Value of the test and p
Region	Yes	No	
North East	576	135	
North West	833	270	
Yorkshire and the Humber	686	201	
East Midlands	624	136	
West Midlands	686	207	
East of England	763	172	
London	674	304	
South East	1130	255	
South West	740	142	

X2 = 98.53
P value = 8.522789e-18

- **Statistical different between males and females based on height.**

- Null hypothesis - there is no significant difference of height between males and female.

- ks.test p value = 0.007038, meaning there is no normality, so because there is no normality I ran a t.test.

- The t-test p value = 1.018285e-05, so is <0.05 meaning we reject the null hypothesis. Therefore there is significant difference in height between males and females.

- **Statistical difference between males and females based on weight.**

- Null hypothesis - there is no significant difference of weight between males and females.

- Ks.test p value = 0.0005736, meaning there is no normality, so because there is no normality I ran a t.test.

- The t-test p value = 9.643029e-07 so is <0.05 meaning we reject the null hypothesis. Therefore there is significant difference in weight between males and females.

Correlation between whether a person drinks nowadays, total household income, age at last birthday and gender.

	Drink nowadays	Total household income	Age at last birthday	Gender
Drink nowadays		P value = 2.220477e-07 R value = -0.05695969	P value = 2.505517e-08 R value = 0.06027989	P value = 6.231612e-27 R value = 0.115942
Total household income	P value = 2.220477e-07 R value = -0.05695969		P value = 5.323502e-14 R value = -0.07403357	P value = 0.008748008 R value = -0.02582917
Age at last birthday	P value = 2.505517e-08 R value = 0.06027989	P value = 5.323502e-14 R value = -0.07403357		P value = 0.001829077 R value = 0.03024415
Gender	P value = 6.231612e-27 R value = 0.115942	P value = 0.008748008 R value = -0.02582917	P value = 0.001829077 R value = 0.03024415	

Interpretation of table on previous slide

- Age at last birthday and drink nowadays has a <0.05 p value and an R value of 0.06 this makes these two variables significant.
- Total household income and drink nowadays has a <0.05 p value and an R value of -0.06 this makes these two variables not significantly different.
- Age at last birthday and total household income has a <0.05 p value and an R value of -0.07 this makes these two variables not significantly different.
- Gender and drink nowadays has a <0.05 p value and an R value of 0.12 this makes these two variables significant.
- Gender and total household income has a <0.05 p value and an R value of -0.03 this makes these two variables not significantly different.
- Gender and age at last birthday has a <0.05 p value and an R value of 0.03 this makes these two variables not significantly different.

Discussion and conclusion

- According to renters for disease control and prevention, women tend to have a higher alcohol level than men after drinking the same amount of alcohol and the effect usually occur more quickly and last longer in women. (CDC, 2024)
- Out of 10617 people in the data, 63.2% of them drank alcohol and 54.2% of them being female.
- The mean age was 42, and the mean BMI was 25.92, this will have an effect on the alcohol in women due to their body fat.
- 3172 males drink alcohol and 605 females drink alcohol out of the whole 10617.
- South east had the highest drinking status at 1130.
- My conclusion is that there is no significance in the variables depending on weather a person drinks or not.

R code

Q1

```
> nrow(spss.data)
[1] 10617
```

Q2

```
> library(dplyr)
> total_dnnow <-nrow(spss.data)
> drink_nowadays <- dplyr::filter(spss.data, dnnow %in% "1")
> total_1 <- nrow(drink_nowadays)
> percent_1 <- (total_1/total_dnnow)*100
> percent_1
[1] 63.21937
```

Q3

```
> total_sex <-nrow(spss.data)
> all_female<-dplyr::filter(spss.data,Sex %in% "2")
> total_female<-nrow(all_female)
> percent_female<-(total_female/total_sex)*100
> percent_female
[1] 54.29971
```

Q4

```
> max(spss.data$topqual3)
<labelled<double>[1]>: (D) Highest Educational Qualification
[1] NA

Labels:
 value                       label
    -9                     Refused
    -8                  Don't know
    -7       Refused/not obtained
    -6     Schedule not obtained
    -2   Schedule not applicable
    -1             Not applicable
     1  NVQ4/NVQ5/Degree or equiv
     2     Higher ed below degree
     3     NVQ3/GCE A Level equiv
     4     NVQ2/GCE O Level equiv
     5 NVQ1/CSE other grade equiv
     6              Foreign/other
     7          No qualification
```

Q5

```
> View(all_divorced)
> total_marital_status <- nrow(spss.data)
> all_divorced<-dplyr::filter(spss.data, marstatc %in% "5")
> total_divorced<-nrow(all_divorced)
> percent_divorced<-(total_divorced/total_marital_status)*100
> percent_divorced
[1] 5.594801
```

Q6

```
> total_marital_status <- nrow(spss.data)
> all_seperated<-dplyr::filter(spss.data, marstatc %in% "4")
> total_seperated<-nrow(all_seperated)
> percent_seperated<-(total_seperated/total_marital_status)*100
> percent_seperated
[1] 2.109824
```

Q7

```
> range(spss.data$HHSize)
[1]  1 10
> range(spss.data$Age)
[1]   0 100
> summary(spss.data$bmival)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   8.34   21.93   25.59   25.92   29.39   65.28    2241
> summary(spss.data$HHSize)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000   2.000   3.000   2.851   4.000  10.000
> summary(spss.data$bmival)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
   8.34   21.93   25.59   25.92   29.39   65.28    2241
> summary(spss.data$Age)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   0.00   22.00   42.00   41.56   61.00  100.00
>
```

```
> table(spss.data$HHSize)

   1    2    3    4    5    6    7    8    9   10
1594 3490 2067 2406  719  218   61   37    9   16
> which.max(table(spss.data$HHSize))
2
2
```

```
 [ reached getOption("max.print") -- omitted 7192 entries ]
> which.max(table(spss.data$bmival))
13.7670587559799
              39
```

```
> sd(spss.data$HHSize)
[1] 1.368528
> sd(spss.data$Age)
[1] 23.83203
> sd(spss.data$bmival, na.rm=TRUE)
[1] 6.138344
```

Q8

```
> test_data = data.frame(spss.data$dnnow, spss.data$Sex)
> test_data = table(spss.data$dnnow, spss.data$Sex)
> print(test_data)

     1    2
  1 3172 3540
  2  605 1217
> chisq.test(test_data)

        Pearson's Chi-squared test with Yates' continuity
        correction

data:  test_data
X-squared = 114.15, df = 1, p-value < 2.2e-16
```

Q9

```
> test_data1 = data.frame(spss.data$dnnow, spss.data$gor1)
> test_data1 = table(spss.data$dnnow, spss.data$gor1)
> print(test_data1)

     1    2    3    4    5    6    7    8    9
  1  576  833  686  624  686  763  674 1130  740
  2  135  270  201  136  207  172  304  255  142
> chisq.test(test_data1)

        Pearson's Chi-squared test

data:  test_data1
X-squared = 98.53, df = 8, p-value < 2.2e-16
```

Q10

```
> female<-spss.data%>%filter(spss.data$Sex==2)
> male<-spss.data%>%filter(spss.data$Sex==1)
> female1<-na.omit(female$htval)
> male1<-na.omit(male$htval)
> tablefemale<-table(female1)
> tablemale<-table(male1)
> ks.test(tablefemale, tablemale)

        Asymptotic two-sample Kolmogorov-Smirnov test

data:  tablefemale and tablemale
D = 0.089352, p-value = 0.007038
alternative hypothesis: two-sided

Warning message:
In ks.test.default(tablefemale, tablemale) :
  p-value will be approximate in the presence of ties
> t.test(tablefemale, tablemale)

        Welch Two Sample t-test

data:  tablefemale and tablemale
t = 4.4319, df = 1217.5, p-value = 1.018e-05
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 1.026068 2.656097
sample estimates:
mean of x mean of y
 7.060921  5.219839

> test<-t.test(tablefemale, tablemale)
> test$p.value
[1] 1.018285e-05
```

Q11

```
> F1<-spss.data%>%filter(spss.data$Sex==2)
> M2<-spss.data%>%filter(spss.data$Sex==1)
> cleanF1<-na.omit(F1$wtval)
> cleanM1<-na.omit(M2$wtval)
> Ftable1<-table(cleanF1)
> Mtable1<-table(cleanM1)
> ks.test(Ftable1, Mtable1)

        Asymptotic two-sample Kolmogorov-Smirnov test

data:  Ftable1 and Mtable1
D = 0.089528, p-value = 0.0005736
alternative hypothesis: two-sided

Warning message:
In ks.test.default(Ftable1, Mtable1) :
  p-value will be approximate in the presence of ties
> t.test(Ftable1, Mtable1)

        Welch Two Sample t-test

data:  Ftable1 and Mtable1
t = 4.9151, df = 1880.5, p-value = 9.643e-07
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 0.4926111 1.1467467
sample estimates:
mean of x mean of y
 4.710711  3.891032

> test<-t.test(Ftable1, Mtable1)
> test$p.value
[1] 9.643029e-07
```

Q12

```
> cor.test(spss.data$Sex, spss.data$dnnow,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$Sex and spss.data$dnnow
S = 9.1577e+10, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
     rho
0.115942

Warning message:
In cor.test.default(spss.data$Sex, spss.data$dnnow, method = "spear
man") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$Sex, spss.data$dnnow,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$Sex, spss.data$dnnow, method = "spear
man") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 6.231612e-27
> data1$estimate
     rho
0.115942
>
```

```
> cor.test(spss.data$Sex, spss.data$Age,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$Sex and spss.data$Age
S = 1.9343e+11, p-value = 0.001829
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.03024415

Warning message:
In cor.test.default(spss.data$Sex, spss.data$Age, method = "spearma
n") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$Sex, spss.data$Age,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$Sex, spss.data$Age, method = "spearma
n") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 0.001829077
> data1$estimate
       rho
0.03024415
>
```

```
> cor.test(spss.data$Age, spss.data$dnnow,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$Age and spss.data$dnnow
S = 9.7343e+10, p-value = 2.506e-08
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
0.06027989

Warning message:
In cor.test.default(spss.data$Age, spss.data$dnnow, method = "spear
man") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$Age, spss.data$dnnow,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$Age, spss.data$dnnow, method = "spear
man") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 2.505517e-08
> data1$estimate
       rho
0.06027989
```

```
> cor.test(spss.data$Age, spss.data$totinc,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$Age and spss.data$totinc
S = 1.9572e+11, p-value = 5.324e-14
alternative hypothesis: true rho is not equal to 0
sample estimates:
        rho
-0.07403357

Warning message:
In cor.test.default(spss.data$Age, spss.data$totinc, method = "spea
rman") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$Age, spss.data$totinc,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$Age, spss.data$totinc, method = "spea
rman") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 5.323502e-14
> data1$estimate
        rho
-0.07403357
>
```

```
> cor.test(spss.data$totinc, spss.data$dnnow,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$totinc and spss.data$dnnow
S = 9.9241e+10, p-value = 2.22e-07
alternative hypothesis: true rho is not equal to 0
sample estimates:
        rho
-0.05695969

Warning message:
In cor.test.default(spss.data$totinc, spss.data$dnnow, method = "sp
earman") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$totinc, spss.data$dnnow,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$totinc, spss.data$dnnow, method = "sp
earman") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 2.220477e-07
> data1$estimate
        rho
-0.05695969
```

```
> cor.test(spss.data$Sex, spss.data$totinc,
+          method = "spearman")

        Spearman's rank correlation rho

data:  spss.data$Sex and spss.data$totinc
S = 1.8693e+11, p-value = 0.008748
alternative hypothesis: true rho is not equal to 0
sample estimates:
        rho
-0.02582917

Warning message:
In cor.test.default(spss.data$Sex, spss.data$totinc, method = "spea
rman") :
  Cannot compute exact p-value with ties
> data1 <- cor.test(spss.data$Sex, spss.data$totinc,
+          method = "spearman")
Warning message:
In cor.test.default(spss.data$Sex, spss.data$totinc, method = "spea
rman") :
  Cannot compute exact p-value with ties
> data1$p.value
[1] 0.008748008
> data1$estimate
        rho
-0.02582917
>
```

References

- CDC, (2024). Excessive alcohol use and women's health. Available at: https://www.cdc.gov/alcohol/fact-sheets/womens-health.htm#:~:text=Alcohol Affects Women Differently than Men&text=After drinking the same amount,longer in women than men. [date accessed 20th July 2024].

- Drink aware, (2024). Alcohol consumption uk. Available at: https://www.drinkaware.co.uk/research/alcohol-facts-and-data/alcohol-consumption-uk [date accessed 20th July 2024].

- NHS England, (2022). Health survey for England, 2021 part 1. Available at: https://digital.nhs.uk/data-and-information/publications/statistical/health-survey-for-england/2021/part-3-drinking-alcohol [date accessed 19th July 2024].

Transcript

Statical analysis presentation by Courtney Sommerville. So in this presentation we talk about the use of alcohol. Alcohol is a widely used beverage and is the leading cause of hospital admissions. Its estimated in 2017-2018 there were 338 thousand admissions just due to alcohol. There were also 5843 deaths due to alcohol. This is 6% higher in 2016 and 16% higher than 2007. 57% of alcohol drank up to 14 units of alcohol each week and 24% drank more than this. 82% of adults drank alcohol in the last 12 months and 49% of adults drink at least once a week. The purpose of this assignment is to be able to use R to look at the data given and perform different tests and to determine if variables are statistically different. Using the data from the health survey of England 2011, I have gathered all the data in a PowerPoint I'm about to show you. I use lots of different analysis during this task this consisted of statistical difference so I performed a t test aswell as finding the values for different questions and also finding mean median and mode, standard deviation and more.

R is used to perform these tests because it's a programming language that allows statistical computing and shows data we are looking for in a large data set. So here we have some data from the data set so we have a total of 10617 people in the sample data, I worked out that 54.2% are female, 63.2% of them drank alcohol, this is the whole data set, there is also 5.59% of the whole dataset divorced, and 2.1% separated. The highest level of education in the sample data is NVQ4 NVQ5 degree or equivalent. So here I calculated the mean median mode minimum maximum range and standard deviation of the variables household size BMI and age at last birthday. Erm so here we have a minimum BMI is 8.34 maximum BMI 65.28 the mean household size is 2.85, the median age at last birthday is 42, and standard deviation for BMI is 6.138. So here we are running a significance test against drinking status and gender. Erm the null hypothesis is there is no relationship between gender and drinking status. Erm so as you can see males drinking status yes is 3172 males drinking status no is 3540. Females drinking status yes is 605 female drinking status no is 1217. So here we have a p value of 1.20834e-26, so the null hypothesis is there is no relationship between gender and drinking status, so I ran a significance test to find out which gender drinks more alcohol and the p value is <0.05 so therefore we reject the null hypothesis as there is a significant difference between gender and drinking. As you can see from the table we can see that the female data is quite much lower than the male data. So here we have a significance test against region and drinking status, so the highest drinking status for yes is in the south east and the lowest drinking status for no is in the north east at 135 and the south east for drinking yes is 1130. So yeah the p value of 8.522789e-18 so we have a null hypothesis of there is no significant difference between drinking and region so I ran a significance test to find out which region drinks more alcohol and the p value is <0.05 so therefore we reject the null hypothesis as there is a significance difference between drinking and the region that they are in. So here we ran a statistical difference between males and females based on height and weight. So on the height one the null hypothesis there is no significance different between height in males and females. So I ran a ks test and the p value came out at 0.007038 meaning there is no normality so because there is no normality I ran a t test and the t test p value came out at 1.018285e-05 so the p value is <0.05 meaning we reject the null hypothesis therefore there is significant difference in height between males and females. The statistical difference between males and females based on weight the null hypothesis is there is no significant difference on weight based on males and females. The ks test p value came out at 0.0005736 meaning there is no normality because of this I ran a t test, the t test value came out at 9.643029e-07 so because the p value is <0.05 meaning we reject the null hypothesis there is significant difference between males and females based on weight. So here is a table where I worked out the correlation between weather a person drinks nowadays, total household income, age at last birthday and gender. Erm so as we can see the p value 2.220477e-07 for total household income by drinking nowadays, for age at last birthday versus drinking nowadays the p value is 2.505517e-08 for age at last birthday versus total household income the p value is 5.323502e-14, for gender versus drinking nowadays the p value is 6.31612e-27 for gender versus total household income the p value of 0.008748008 and for gender and age at last birthday the p value is 0.001829077. So for age at last birthday and drink nowadays because the p value is <0.05 and the R value is 0.06 this makes these two variables significant. Total household income and drinking nowadays has a <0.05 p value and R value is -0.06 this makes these two variables not significantly different. Age at last birthday and total household income has a p value of <0.05 and a R value of -0.07 this makes these two variables not significantly different. Gender and drink nowadays has a p value of <0.05 and a R value of 0.12 this makes these two variables significant. Gender and total household income has a p value of <0.05 and R value of -0.03 this makes these two variables not significantly different. And gender and age at last birthday has a p value of <0.05 and a R value of 0.03 this makes these two variables not significantly different. So this is the interpretation of the table on the previous slide that we just looked at. So according to renters for disease control and prevention women tend to have a higher alcohol level than men after drinking the same amount of alcohol and the effect usually occurs more quickly and lasts longer in women. Out of the 10617 people in the data 63.2% of them drank alcohol and 54.2% being female. The mean age was 42 and the mean BMI was 25.92. This will have an effect on the alcohol in women due to their body fat with how much alcohol they consume and how quickly it affects them or their alcohol level. 3172 males drink alcohol and 605 drink alcohol out of the whole 10617. The south east had the highest drinking status at 1130.

My conclusion is that there is no significant difference in the variables depending of whether a person drinks or not. Erm and then we just have the code I used to perform all of the tests, erm with the questions by it, erm and then references at the end. Erm ok that's all thank you for listening.